🎙️ BeagleBone Black Voice Demo (eSpeak)

🎯 Goal

- Use **eSpeak** on a BeagleBone Black
- Output **audible speech**
- Keep **DNS + Security Onion untouched**
- Demonstrate **edge-device voice alerts**

🧰 Hardware Required

Item	Notes
BeagleBone Black	Debian image
USB audio adapter	BBB has no native audio out
Speaker / headphones	3.5mm or USB
USB power	Stable power is important

⚠️ Do **not** rely on HDMI audio — USB audio is far more reliable.

🖥️ OS Assumption

- **Debian (official BeagleBone image)**
- Logged in via SSH or local console
- Network access allowed (non-infrastructure node)

1️⃣ Update & Install Packages

```
sudo apt update
sudo apt install -y espeak espeak-data alsa-utils
```

2️⃣ Confirm Audio Device

```
aplay -l
```

Expected output example:

```
card 1: Device [USB Audio Device], device 0: USB Audio
```

If you don't see a USB device → unplug/replug adapter.

3️⃣ Set USB Audio as Default

```
nano ~/.asoundrc
```

Paste:

```
defaults.pcm.card 1
defaults.ctl.card 1
```

Save & exit.

Restart ALSA:

```
sudo alsa force-reload
```

4️⃣ Test Raw Audio (Important)

```
speaker-test -t wav
```

✅ You should hear white noise
❌ If not, fix audio before continuing

5️⃣ Test eSpeak (Basic)

```
espeak "BeagleBone Black voice system online"
```

If quiet or silent, use the **reliable method**:

```
espeak "Edge device voice test successful" --stdout | aplay
```

6️⃣ Voice Tuning (Demo Friendly)

```
# Slower, clearer
```

```
espeak -s 140 "This is a clear voice demo"

# Higher pitch
espeak -p 70 "Higher pitch alert"

# Different accent
espeak -v en-sc "Scottish system alert"
```

List voices:

```
espeak --voices
```

7️⃣ Python Voice Alert Demo (🔥 This is the good part)

Create demo script:

```
nano voice_alert.py
import os
import time

def speak(msg):
    os.system(f'espeak "{msg}" --stdout | aplay')

speak("BeagleBone Black alert system initialized")

time.sleep(2)
speak("Motion detected on perimeter sensor")

time.sleep(2)
speak("Security status nominal")
```

Run it:

```
python3 voice_alert.py
```

8️⃣ (Optional) GPIO → Voice Alert Concept

Example logic (pseudo-lab):

```
if motion_detected:
    speak("Warning. Motion detected.")
```

This pairs **perfectly** with:

- PIR sensors
- Door switches
- Lab alerts

- IoT demos

🔐 Security Justification (GitHub-Ready)

- ✔ No DNS changes
- ✔ No pfSense modification
- ✔ No SOC blind spots
- ✔ Edge device only
- ✔ Voice logic isolated from infrastructure

This is **exactly how speech belongs in a secure range**.

🧪 Why BeagleBone Black is Ideal for This

Reason	Benefit
Separate hardware	No infra risk
GPIO native	Sensors + alerts
Debian-based	Easy tooling
Low power	Always-on alerts
Visual demo	Great for YouTube

✅ Demo Talking Point (YouTube / Portfolio)

"In this lab, we intentionally avoided deploying speech services on DNS infrastructure.
Voice alerts were offloaded to an edge device to preserve SOC integrity."

That's **professional-grade reasoning**.

Next (if you want)

I can:

- Add **systemd auto-start** for voice alerts
- Tie BBB alerts to **HTTP / MQTT**

- Forward alerts as **logs to Security Onion**
- Turn this into a **CTF-style audible alert**
- Write a **README + diagram** for GitHub

Just tell me the next step 👌